Trading statistical conservatism for speed: A 'greedy' modification to Boruta

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Abstract. The Boruta algorithm is the most widely adopted method for all-relevant feature selection, but its computational cost and lack of convergence guarantees limit its applicability in modern high-dimensional settings, where the number of features p frequently exceeds 10^4. We propose Greedy Boruta, a modification that replaces Vanilla Boruta's binomial confirmation test with a single-hit rule: a feature is confirmed if it ever exceeds the maximum shadow importance in any iteration. Despite its simplicity, this change yields four formal guarantees: convergence in exactly $K = \lceil \log_2(p/\alpha) \rceil$ iterations, feature recall at least as high as Vanilla Boruta when both methods observe the same importance scores and use the same rejection test, simplification of the final BH rejection calculation and a per-feature false positive rate bounded above by $K/(p+1)$, which decreases as dimensionality grows. The deterministic iteration count eliminates all termination-related hyperparameters and enables embarrassingly parallel execution, implemented in `greedyboruta` (available on PyPI). We further introduce a weak/strong confirmation framework that distinguishes features satisfying the original Boruta confirmation criterion from those selected exclusively by the greedy criterion, providing users with an interpretable measure of confidence among the confirmed features. We also show that the number of shadow features can be reduced from p to $r \leq p$, yielding an explicit trade-off between computational cost and false positive control. On a synthetic benchmark with known ground truth and six real-world high-dimensional datasets, Greedy Boruta achieves 1.8 to 66.2 times speedups while detecting no statistically significant difference in downstream model performance relative to Vanilla Boruta in 15 of the 18 benchmark configurations.

Keywords: Feature selection · Boruta algorithm · All-relevant features · Computational efficiency · Parallelization

1 Introduction

Feature selection is a critical preprocessing step in high-dimensional machine learning pipelines, reducing computational cost, improving model interpretability and mitigating overfitting. Methods for feature selection broadly fall into two paradigms. *Minimal-optimal* methods, such as Lasso [**?**], Ridge [**?**], Recursive Feature Elimination [**?**] and mRMR [**?**], seek the smallest subset of features that maximizes predictive performance. *All-relevant* methods instead aim to

identify every feature that carries information about the target variable, regardless of redundancy. The all-relevant paradigm is particularly valuable in scientific discovery [**?**], biomarker identification [**?**] and causal inference [**?**], where discarding a relevant feature represents a permanent loss of information that no downstream analysis can recover, even if there is redundancy between features.

The Boruta algorithm [**?**,**?**] is the most widely adopted method for all-relevant feature selection. Its statistical framework, based on comparing each feature's importance against permuted "shadow" copies that represent the maximum achievable importance by random noise, provides a principled mechanism for distinguishing signal from noise. Since its introduction, Boruta has been adopted across diverse domains including bioinformatics [**?**,**?**], finance [**?**,**?**] and natural language processing [**?**,**?**], and several variants have been proposed. For instance, BorutaPy [**?**] offers an equivalent Python implementation to the original R implementation, while BorutaSHAP [**?**] replaces impurity-based importance with SHAP-value [**?**] based importances. However, none of the available variants address two fundamental limitations of the algorithm.

First, Boruta is computationally expensive. Each iteration requires training a tree-based ensemble on a dataset augmented with p shadow features, and the algorithm may require hundreds or thousands of iterations to resolve all features. In modern industrial and scientific applications where p is often in the thousands, running Boruta to completion can require days of computation, making the method impractical in time-constrained settings. Second, Boruta provides no convergence guarantees (the state in which all features are either confirmed or rejected). That is, features can remain *tentative*, neither confirmed nor rejected, forcing practitioners to either increase the iteration budget arbitrarily, enable heuristic early stopping mechanisms or resolve remaining tentatives using post-hoc heuristics such as the rough fix [**?**]. In practice, this uncertainty requires careful tuning of hyperparameters that control the maximum number of iterations and early-stopping, creating a tension between computational cost and quality of the solution. etermines the tota

In this paper, we propose Greedy Boruta, a modification that replaces Boruta's binomial-test confirmation criterion with a single-hit rule: a feature is confirmed if it exceeds the maximum shadow importance in at least one iteration. This change is motivated by the fact that, under the all-relevant paradigm, false negatives (missed relevant features) are often more costly than false positives (incorrectly included noise features), since false positives can be filtered by downstream regularization, cross-validation, stability analyses or minimal-optimal feature selection procedures. We further introduce two practical extensions. First, we partition confirmed features into *strongly confirmed* and *weakly confirmed* groups. Strongly confirmed features satisfy the original Boruta confirmation criterion and therefore would also be selected by Vanilla Boruta, whereas weakly confirmed features are selected exclusively by Greedy Boruta and may be viewed as additional candidate variables for further investigation. Second, we show that the number of shadow features can be reduced from p to $r \leq p$, substantially decreasing the dimensionality of the augmented dataset and yielding further

computational savings in high-dimensional settings. Throughout this paper, we refer to the original algorithm and its equivalent reimplementations collectively as *Vanilla Boruta*, and to our proposed modification as *Greedy Boruta*.

The main contributions of this paper are:

1. **Algorithmic:** A modification to Boruta that guarantees convergence in exactly $K = \lceil \log_2(p/\alpha) \rceil$ iterations[1], eliminating all termination-related hyperparameters. We further introduce a weak/strong confirmation framework that distinguishes features satisfying the original Boruta confirmation criterion from those selected exclusively by the greedy criterion.
2. **Theoretical:** Proofs of (i) a recall guarantee showing that the set of selected features by Vanilla Boruta is contained in the set of features selected by Greedy Boruta (Theorem **??**), (ii) a convergence bound on the number of iterations (Theorem **??**), (iii) simplification of the final Benjamini–Hochberg rejection calculation (Corollary **??**), (iv) a false positive rate (FPR) upper bound of $K/(p+1)$ (Theorem **??**) and (v) an analysis of the effect of reducing the number of shadow features from p to $r \leq p$ on the computational and statistical properties of Greedy Boruta.
3. **Practical:** A reduced-shadow variant of Greedy Boruta that decreases the dimensionality of the augmented dataset from $2p$ to $p + r$, yielding additional computational savings while allowing practitioners to explicitly control the trade-off between runtime and false positive rate. The deterministic iteration count and independence across iterations enable embarrassingly parallel execution, implemented in the `greedyboruta` PyPI package.
4. **Practical:** The `greedyboruta` package, available on PyPI, provides a Python implementation of Greedy Boruta. This implementation includes a reduced-shadow parameter that uses $r \leq p$ shadow features, reducing the dimensionality of the augmented dataset from $2p$ to $p + r$ and offering a tunable trade-off between computational cost and the bound on the per-feature false positive rate.

2 Background: The Boruta Algorithm

The idea behind Vanilla Boruta is to construct *shadow features*, permuted copies of features that serve as competitive benchmarks to the non-shadow features. By comparing each non-shadow feature's importance against the maximum shadow importance across multiple randomized trials, Boruta makes statistically principled decisions about which features carry genuine signal and which are not statistically distinguishable from noise.

The algorithm operates iteratively, with each iteration consisting of four steps.

1. **Shadow feature construction**: for each of the p original features, Boruta creates a shadow copy by randomly permuting the feature's values across samples. This permutation destroys any relationship with the target variable

[1] Under Bonferroni correction.

while preserving the feature's marginal distribution. The dataset is then extended to include both the p original features and p shadow features.
2. **Importance scoring**: a tree-based ensemble (originally random forests [**?**]) is trained on this extended dataset, and feature importance scores are extracted for all $2p$ features.
3. **Hit registration**: each original feature whose importance exceeds the maximum shadow importance in that iteration receives a 'hit', and these hits are accumulated across iterations.
4. **Statistical testing**: binomial tests determine whether the features are confirmed (relevant), tentative (unknown) or rejected (irrelevant).

The statistical testing framework employs two one-sided binomial tests at each iteration. For a feature with h hits across K iterations, the confirmation test checks whether $\mathbb{P}(X \geq h \mid K, \theta = 0.5) \leq \alpha$, where α denotes statistical significance and $p = 0.5$ represents the null hypothesis that the feature is no better than random noise (represented by the best shadow). The symmetric rejection test checks $\mathbb{P}(X \leq h \mid K, \theta = 0.5) \leq \alpha$. Both tests are typically applied with a feature-wise Bonferroni correction to control the family-wise error rate across the p features under test. Features passing neither test remain tentative. In both the R and Python implementations, these tests are repeated at every iteration (in `boruta_py`, from the fifth iteration onward) and decisions are final: only tentative features are re-tested, and rejected features are removed from all subsequent iterations.

Because there are no convergence guarantees with Vanilla Boruta, when the algorithm terminates with tentative features unresolved, whether by early stopping or reaching a maximum number of iterations, features that remain tentative are resolved using the rough fix heuristic [**?**]. Under this heuristic, a tentative feature is confirmed if its median importance across iterations exceeds the median of the per-iteration maximum shadow importances, and rejected otherwise.

The use of artificial variables as negative controls also underlies knockoff methods [**?**,**?**]. However, model-X knockoffs are designed to control the false discovery rate, whereas Boruta targets all-relevant feature selection through comparisons with permuted shadow features. Knockoff guarantees also rely on an exchangeability property of the joint distribution of original and knockoff covariates that permuted shadow features do not generally satisfy; the exchangeability assumption used in our analysis (Section **??**) concerns only the importance scores. Another related approach, PIMP [**?**], calibrates feature importance against null distributions estimated by repeatedly permuting the response.

3 The Greedy Boruta Algorithm

Motivated by the asymmetry between false negatives and false positives discussed in Section **??**, Greedy Boruta replaces Vanilla Boruta's binomial confirmation test with a single-hit rule: a feature is confirmed if its importance exceeds the

maximum shadow importance in at least one iteration. The rejection criterion is retained, but "evaluated" only after a predetermined number of iterations.

Algorithm **??** presents the complete Greedy Boruta procedure. Given a dataset with p features and a target variable y, the algorithm first determines the total required number of iterations K, which depends on the number of features and significance level. At each iteration, shadow features are generated by randomly permuting the values of every original feature, thereby preserving marginal distributions while destroying any association with the target. A tree-based ensemble model is then trained on the augmented dataset containing both original and shadow features, and feature importance scores are computed. The maximum importance score among the shadow features defines a noise threshold for that iteration. Any original feature whose importance exceeds this threshold receives a hit. After all iterations are completed, features with at least one hit are confirmed as relevant, whereas features with zero hits are rejected.

The key difference relative to Vanilla Boruta lies in the confirmation step, while the rejection criterion remains effectively unchanged. Vanilla Boruta employs sequential binomial hypothesis tests to determine whether a feature should be confirmed or rejected, whereas Greedy Boruta confirms any feature that achieves at least one hit across all iterations. This modification appears in lines 14 and 15 of Algorithm **??**, where the greedy confirmation rule replaces the statistical confirmation test. Furthermore, because a feature with zero hits never exceeds the maximum shadow importance in any iteration, rejection can be determined directly at the end of the procedure without repeatedly performing intermediate statistical tests. Consequently, Greedy Boruta eliminates the sequential testing mechanism used in Vanilla Boruta, resulting in a simpler and more computationally efficient algorithm while preserving the same underlying principle of comparing real features against randomized shadow features.

3.1 Weak and strong confirmations

A potential limitation of the Greedy Boruta confirmation rule is that all features with at least one hit are treated equally, regardless of the total number of hits accumulated across iterations. In particular, a feature that beats the shadow threshold once receives the same confirmation status as a feature that consistently outperforms the shadow threshold. To provide a more nuanced interpretation of the selected features while preserving the feature recall guarantee established in Theorem **??**, which is presented in Section **??**, we propose partitioning the confirmed features into *strongly* and *weakly* confirmed groups.

Let H_j denote the number of hits accumulated by feature j after K iterations. Recall that Greedy Boruta confirms a feature whenever $H_j \geq 1$. Define the set of confirmed features as

$$\mathcal{C}_G = \{j : H_j \geq 1\}.$$

We further define the set of *strongly confirmed* features as

$$\mathcal{C}_{\text{strong}} = \{j : \mathbb{P}(X \geq H_j) \leq \alpha/p, \quad X \sim \mathsf{Binomial}(K, \theta = 0.5)\}\,.$$

Algorithm 1 Greedy Boruta Algorithm

Require: Dataset $X \in \mathbb{R}^{n \times p}$, target y, significance level α
Ensure: Set of confirmed features $\mathcal{C}$, rejected features $\mathcal{R}$
Compute $K = \lceil \log_2(p/\alpha) \rceil$ {Total iterations}
Initialize hits$[j] = 0$ for all features $j = 1, \ldots, p$
for $i = 1$ to K **do**
  Create shadow features by permuting each column of X
  Train tree ensemble on $[X, X_{\text{shadow}}]$ with target y
  Compute importance scores for all features
  $m \leftarrow \max$(importance of shadow features)
  for each original feature j **do**
    if importance$[j] > m$ **then**
      hits$[j] \leftarrow$ hits$[j] + 1$
    end if
  end for
end for
$\mathcal{C} \leftarrow \{j : \text{hits}[j] \geq 1\}$ {Greedy confirmation}
$\mathcal{R} \leftarrow \{j : \text{hits}[j] = 0\}$ {All others rejected (p-value $\leq \alpha/p$)}
return $\mathcal{C}, \mathcal{R}$

That is, a feature is strongly confirmed if it satisfies the original Boruta confirmation criterion under the Bonferroni correction. Equivalently, these are the features that would be confirmed by Vanilla Boruta after K iterations.

The remaining confirmed features are classified as *weakly confirmed* and are part of the set

$$\mathcal{C}_{\text{weak}} = \{j : H_j \geq 1 \text{ and } \mathbb{P}(X \geq H_j) > \alpha/p, \quad X \sim \mathsf{Binomial}(K, \theta = 0.5)\}.$$

The distinction between weak and strong confirmations provides a natural ranking of the selected features. Strongly confirmed features have accumulated sufficient evidence to satisfy the original Boruta significance criterion and therefore correspond to features that would also be selected by Vanilla Boruta. In contrast, weakly confirmed features are selected exclusively by Greedy Boruta. These features have demonstrated some evidence of relevance by outperforming the shadow threshold at least once, but the evidence is not strong enough to achieve statistical significance under the Boruta binomial test.

From a practical perspective, strongly confirmed features may be interpreted as high-confidence discoveries, whereas weakly confirmed features may warrant additional investigation through domain expertise, stability analyses or downstream predictive validation. This distinction allows Greedy Boruta to retain its computational advantages and feature recall guarantee while providing users with a measure of confidence among the selected features.

3.2 Theoretical Properties

We now state the theoretical guarantees of Greedy Boruta. Full proofs are provided in Appendix **??**.

Feature Recall Guarantee Our first result shows that relaxing the confirmation criterion never costs recall relative to Vanilla Boruta.

Theorem 1 (Feature Recall Guarantee). *Suppose Vanilla Boruta, as described in Section* **??**, *and Greedy Boruta use the same significance level and correction setting and observe the same feature and maximum-shadow importance scores through iteration K, and that Vanilla Boruta runs for at least K iterations. Let $\mathcal{C}_V$ and $\mathcal{C}_G$ denote the sets of features they confirm. Then $\mathcal{C}_V \subseteq \mathcal{C}_G$, whether Vanilla Boruta confirms a feature through the binomial test or through the rough fix.*

The result follows because Greedy Boruta's confirmation rule is the weakest possible: a single hit. A feature with no hits in the first K iterations cannot be confirmed by Vanilla Boruta's binomial test, since its confirmation p-value is 1, and because Vanilla Boruta applies the same rejection test at iteration K, it is rejected there. Because rejections are final, the rough fix, which only acts on tentative features, cannot confirm it later either. Every feature Vanilla Boruta confirms therefore has at least one hit by iteration K, which is exactly Greedy Boruta's confirmation rule. Consequently, Greedy Boruta's feature recall is never lower than Vanilla Boruta's.

Convergence Guarantee Unlike Vanilla Boruta, whose number of required iterations cannot be known in advance, Greedy Boruta runs for a fixed number of iterations that is sufficient to resolve every feature.

Theorem 2 (Convergence Guarantee). *Consider a dataset with p features, and let α be the significance level of the binomial rejection test. With Bonferroni correction, after $K = \lceil \log_2(p/\alpha) \rceil$ iterations of Algorithm* **??**, *every feature is either confirmed or rejected, so no feature remains tentative.*

At iteration K, every feature with at least 1 hit is confirmed by the greedy criterion. Every remaining feature has zero hits and thus a rejection p-value of $(0.5)^K$. Under Bonferroni correction, rejection requires $(0.5)^K \leq \alpha/p$, which holds by construction of $K = \lceil \log_2(p/\alpha) \rceil$. Therefore, all features are classified and the algorithm terminates. It is worth noting that the number of iterations decomposes as $K = \lceil \log_2(p) + \log_2(1/\alpha) \rceil$: halving α or doubling p adds exactly one iteration. Table **??** shows K for representative values.

Table 1: Number of iterations $K = \lceil \log_2(p/\alpha) \rceil$ required for convergence when using Bonferroni correction for multiple comparisons.

	Number of features p				
α	50	100	500	1000	5000
0.100	9	10	13	14	16
0.050	10	11	14	15	17
0.010	13	14	16	17	19
0.001	16	17	19	20	23
10^{-4}	19	20	23	24	26
10^{-5}	23	24	26	27	29

Without Bonferroni correction (analogous to `mcAdj=FALSE` in the R implementation), the iteration count reduces to $K = \lceil \log_2(1/\alpha) \rceil$, and the number of iterations is not dependent on the number of features anymore. In Sections **??** and **??**, we use the Bonferroni correction to obtain K.

The same bound also limits Vanilla Boruta: with fewer than K iterations, no binomial test can reach significance, so Vanilla Boruta cannot confirm or reject any feature before iteration K. Greedy Boruta therefore finishes at the earliest iteration at which Vanilla Boruta can make its first decision. Because the iteration count is known in advance, the termination-related parameters of `boruta_py` (`max_iter`, `early_stopping` and `n_iter_no_change`) are no longer needed.

Parallel execution. Because the iteration budget K is fixed in advance (Theorem **??**) and no features are removed during execution, every iteration uses the same feature set and differs only in its random seed. The iterations are therefore independent: each produces a vector of hits, and the final hit counts are their sum. All K iterations can run in parallel, and their results only need to be combined at the final classification step. Vanilla Boruta cannot be parallelized this way, because each iteration depends on the previous ones: rejected features are dropped, the binomial tests are re-evaluated after every iteration and early stopping tracks consecutive iterations without change. `greedyboruta` implements this iteration-level parallelism with `joblib`; its practical benefit depends on the base estimator's own parallelism, as discussed in Section **??**.

3.3 Multiple Testing Corrections

In Vanilla Boruta, multiple-testing correction is required both across features and across iterations, because confirmation and rejection decisions are revisited at every iteration. Greedy Boruta performs a single rejection test at the final iteration and does not use a hypothesis test for confirmation, so trial-wise correction becomes unnecessary. The following corollary of Theorem **??** shows that FDR correction becomes unnecessary as well.

Corollary 1 (BH Rejection Simplification). *At iteration K, every feature not confirmed by Greedy Boruta has zero hits and therefore the same rejection*

Table 2: Mapping between correction settings across implementations.

`greedyboruta`	R Boruta	`boruta_py`
`bonferroni=True`	`mcAdj=TRUE`	`two_step=False`
`bonferroni=False`	`mcAdj=FALSE`	`two_step=True`[2]

p-value 2^{-K}*. The Benjamini–Hochberg procedure [?] applied to these features rejects all of them, exactly as the uncorrected test does.*

Applied to a set of identical p-values, the Benjamini–Hochberg procedure reduces to comparing their common value with α, and $2^{-K} \leq \alpha$ holds under either correction setting. In Vanilla Boruta, by contrast, tentative features have different hit counts and therefore different p-values, so FDR correction is not trivial there. Note that Corollary **??** concerns only the rejection of zero-hit features. It does not establish FDR control among the confirmed features, since greedy confirmation bypasses this test; false positives among confirmed features are analysed in Section **??**. As a result, the only non-trivial correction remaining in Greedy Boruta is the feature-wise Bonferroni adjustment, controlled through the `bonferroni` parameter. Table **??** maps this setting to its equivalents in the most widely used Boruta implementations.

3.4 False Positive Rate Analysis

The feature recall and convergence guarantees come at a cost: by relaxing the confirmation criterion, Greedy Boruta may confirm features that Vanilla Boruta would leave tentative or reject. We quantify this trade-off under a standard assumption of the Boruta framework.

Assumption 1 (Shadow Exchangeability)**.** *An irrelevant feature is exchangeable with the shadow features with respect to the importance measure. That is, the importance of an irrelevant feature and the importances of the p shadow features are identically distributed and symmetric under permutation within each iteration.*

Violations of this assumption (e.g., importance measure biases toward high-cardinality features [**?**,**?**]) can affect both Vanilla Boruta and Greedy Boruta, since both use the same shadow mechanism. Their different decision rules can nevertheless respond differently to these violations. The analysis below isolates the effect of the modified decision rule under the stated assumption.

Under the shadow exchangeability assumption, the importance of an irrelevant feature and the p shadow importances form $p+1$ exchangeable random variables. Each has the same probability of being the unique maximum, so the per-iteration hit probability satisfies $\theta_0 \leq 1/(p+1)$. Equality holds when the maximum is unique almost surely; ties can reduce the probability of strictly exceeding all shadows.

[2] The `two_step=True` mode applies both FDR and trial-wise Bonferroni correction; both degenerate in Greedy Boruta (Corollary **??**), reducing to the uncorrected case.

Table 3: Upper bound on per-feature false positive rate (FPR) of Greedy Boruta under Bonferroni correction ($K = \lceil \log_2(p/\alpha) \rceil$). Each cell shows the value of the bound in Equation (**??**).

	$p = 100$		$p = 500$		$p = 1{,}000$		$p = 10{,}000$	
α	K	FPR	K	FPR	K	FPR	K	FPR
0.05	11	10.89%	14	2.79%	15	1.50%	18	0.18%
0.01	14	13.86%	16	3.19%	17	1.70%	20	0.20%
0.001	17	16.83%	19	3.79%	20	2.00%	24	0.24%
0.0001	20	19.80%	23	4.59%	24	2.40%	27	0.27%
0.00001	24	23.76%	26	5.19%	27	2.70%	30	0.30%

Theorem 3 (FPR of Greedy Boruta). *Suppose Assumption* **??** *holds in every iteration, and let* $K = \lceil \log_2(p/\alpha) \rceil$*. Then, for any irrelevant feature* j*,*

$$\mathbb{P}(j \text{ is confirmed by Greedy Boruta}) \leq \frac{K}{p+1} = \frac{\lceil \log_2(p/\alpha) \rceil}{p+1}. \tag{1}$$

Consequently, if m_0 *of the* p *features are irrelevant, the expected number of false positives satisfies* $\mathbb{E}[\mathrm{FP}] \leq m_0 K/(p+1)$*.*

The proof, given in Appendix **??**, applies a union bound over the K iterations: an irrelevant feature is falsely confirmed if it obtains a hit in at least one iteration, and each iteration contributes a hit probability of at most $1/(p+1)$. The bound on $\mathbb{E}[\mathrm{FP}]$ then follows by summing over the irrelevant features (linearity of expectation). Neither step requires independence, so the result holds even when iterations are dependent or irrelevant features are correlated. Note that $\mathbb{E}[\mathrm{FP}]$ counts false positives; it does not bound their proportion among the selected features (the FDR), which also depends on how many relevant features are confirmed.

The bound is also close to the exact value in an idealized setting. If hits in different iterations were independent and ties impossible, an irrelevant feature would miss in all K iterations with probability $(p/(p+1))^K$, so it would be falsely confirmed with probability exactly $1 - (p/(p+1))^K$. For $p = 500$ and $\alpha = 0.05$, this is 2.76%, against the bound of 2.79%.

The bound also has a favorable structure: the numerator grows logarithmically while the denominator grows linearly, so the FPR *decreases* as p increases. Table **??** presents the bound for representative configurations. For moderate to large p (the typical Boruta use case), the bound is small: at most 5.2% for $p = 500$ and below 3% for $p \geq 1{,}000$ across all α values shown. For small p (e.g., $p = 100$), the bound can exceed 10%, but in such settings the computational cost of Vanilla Boruta is typically manageable.

The same analysis naturally extends to the case where only $r \leq p$ shadow features are generated at each iteration. Under Assumption **??**, the per-iteration hit probability of an irrelevant feature becomes $\theta_0^\star \leq 1/(r+1)$, yielding the corresponding upper bound $\mathbb{P}(\mathrm{FP}) \leq K/(r+1)$. Although reducing the number

of shadow features weakens false positive control, it decreases the dimensionality of the augmented dataset from $2p$ to $p + r$, providing additional computational and memory savings. This reduction in shadow dimensionality may be particularly relevant in distributed computing environments such as Apache Spark [**?**], where constructing and materializing shadow features may become a bottleneck. A detailed theoretical analysis of this reduced-shadow variant, together with empirical results illustrating the resulting computational–statistical trade-off, is presented in Appendix **??**.

4 Experimental Evaluation

The experiments test the claims of Section **??** in turn. Section **??** examines convergence and computational cost (Theorem **??** and parallel execution): whether Vanilla Boruta resolves all features in practice, where its runtime goes, and how much of it Greedy Boruta avoids. Section **??** uses synthetic data with known relevant features to measure what the single-hit rule costs in recall and false positives (Theorems **??** and **??**) and whether the weak/strong split separates them. Section **??** asks, on six real high-dimensional datasets, which features Greedy Boruta adds and whether a downstream model is affected. All experiments in this section use p shadow features; the reduced-shadow variant is evaluated in Appendix **??**.

4.1 Experimental Setup

Datasets. The synthetic benchmark draws $n = 1{,}000$ samples of $p = 500$ independent standard normal features. Five *direct* predictors generate a binary target through a logistic model with signal strength 2, and 20 *proxies* are noisy copies of the direct predictors (correlation 0.8); in line with the all-relevant paradigm, the 25 direct and proxy features form the relevant set and the remaining $m_0 = 475$ are pure noise. Three variants change one factor at a time: $p = 2{,}000$ (dimensionality), signal strength 0.5 (weak signal) and a *null* scenario in which the target is independent of all 500 features, so that every selected feature is a false positive. The real-world benchmark comprises six public high-dimensional datasets (Table **??**): Madelon, Isolet, Gisette, Arcene and Dexter (classification) and Riboflavin (regression) [**?**,**?**,**?**].

Protocol. Every dataset is evaluated with three repetitions of shuffled five-fold cross-validation (15 folds; stratified for classification). All preprocessing and feature selection are fitted on the training part of each fold only, and all methods see exactly the same splits. A downstream LightGBM model [**?**] (256 trees, maximum depth 5, learning rate 0.1) is trained on the selected features and scored on the held-out part by weighted F1 (classification) or R^2 (regression); an empty selection is scored with a training-set constant predictor. As a reference we also report the same model trained on *all features*. The significance level is swept over $\alpha \in \{10^{-1}, 10^{-3}, 10^{-5}\}$; unless stated otherwise, figures use $\alpha = 10^{-3}$, and Section **??** shows that conclusions do not depend on this choice.

Methods. Both methods use `ExtraTrees` [**?**,**?**] with maximum depth 5 and Boruta's automatic tree count. Vanilla Boruta (`boruta_py` with `two_step=False`, matching R's default `mcAdj=TRUE`) runs for at most 512 iterations with early stopping after 128 iterations without a decision. We report both its strict output (confirmed features only) and its rough-fix output (confirmed plus tentative features resolved by the rough fix); the rough fix is the comparison reference, since it is the only Vanilla output that resolves every feature. Greedy Boruta uses Bonferroni correction and the same seeds. To isolate the decision rule from Monte Carlo noise, we additionally evaluate a *coupled* Greedy Boruta: the greedy rule applied to the first K iterations of the very Vanilla run it is compared with. Because no feature can be confirmed or rejected before iteration K (Section **??**), these iterations are identical to Greedy Boruta's computation, so the coupled pair satisfies the conditions of Theorem **??** exactly.

Parallelism and hardware. All runs share a budget of 28 threads on an Intel Core i7-14700K (20 cores, 28 threads) with 64 GB of RAM. Vanilla Boruta and serial Greedy Boruta give all 28 threads to tree construction; parallel Greedy Boruta runs 7 iterations concurrently with 4 threads each. Serial and parallel Greedy Boruta select identical features in all 495 paired fits, so parallelism affects runtime only.

Statistical analysis. For each dataset and α we compute the paired fold-wise difference $D = \text{score}_{\text{Greedy}} - \text{score}_{\text{Vanilla}}$ and test non-inferiority, $H_0 \colon \mathbb{E}[D] \le -\delta$ against $H_1 \colon \mathbb{E}[D] > -\delta$, with a margin $\delta = 0.01$ (weighted F1 and R^2) fixed before the experiment. Because the 15 folds share training data, the standard error uses the Nadeau–Bengio correction [**?**], $\sqrt{(1/15 + n_{\text{test}}/n_{\text{train}})\, s_D^2}$, with 14 degrees of freedom. Tests are run at one-sided level 0.01 with Holm correction over the full prespecified family of 165 comparisons; as a sensitivity analysis we also report the conventional one-sided level 0.025. Failure to establish non-inferiority is inconclusive, not evidence of inferiority.

4.2 Convergence and Computational Cost

Vanilla Boruta rarely converges. On Arcene and Gisette, Vanilla Boruta reached the 512-iteration cap in every fold and at every α; on Dexter and Isolet it did so in 47–93% of folds. In every fold of Arcene, Dexter, Gisette and Isolet, at every α, its final output depended on the rough fix to resolve features the binomial tests left tentative (up to 119 features in a single fold). Greedy Boruta, by construction, resolved every feature after $K = 13$–31 iterations.

Where Vanilla's runtime goes. Fig. **??** tracks complete Vanilla runs, refitted with per-iteration logging (three folds per dataset; the refits reproduce the logged selections exactly). No feature can be decided before iteration K; at K, Vanilla rejects every feature without a hit and confirms the strongly supported ones, leaving only 0.6% (Dexter) to 15% (Gisette) of the features tentative. Reaching this point costs between 4% (Isolet) and 65% (Riboflavin) of Vanilla's total

runtime. The remaining iterations, typically hundreds, are spent on this small undecided remainder, and 0.1–3.4% of features are still tentative when Vanilla stops.

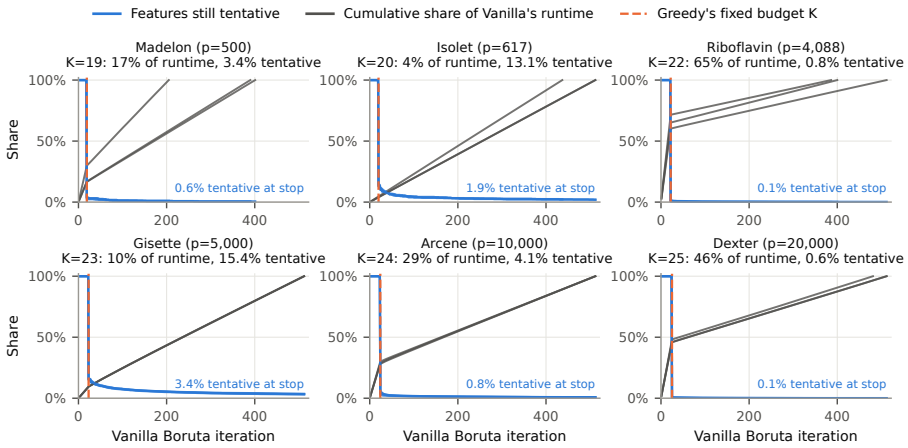


Fig. 1: Anatomy of Vanilla Boruta runs ($\alpha = 10^{-3}$, three folds per dataset). Blue: share of features still tentative; gray: cumulative share of the run's wall-clock time; dashed: Greedy Boruta's fixed budget K. Panel titles give both quantities at iteration K.

Speedup. This share determines the speedup (Fig. **??**). The ratio of iterations, 12.8–39.4 across datasets and α, overstates the gain: Vanilla's later iterations are much cheaper than its first K, because rejected features leave the augmented matrix and the automatic tree count shrinks with them, whereas Greedy Boruta keeps all p features in every iteration. Stopping Vanilla's own computation at iteration K would yield 1.5× (Riboflavin) to 25× (Isolet), and serial Greedy Boruta achieves exactly this: its observed speedup is within 4% of that bound on five of the six datasets (Madelon: 4.6× against 5.9×). Iteration-level parallelism adds a further 1.1–3.9×, for median speedups of 1.7× (Riboflavin) to 57× (Isolet) at $\alpha = 10^{-3}$, and up to 80× at $\alpha = 10^{-1}$. Serial Greedy Boruta also uses 1.1–25× fewer CPU-seconds than Vanilla, so its gain is a reduction in work and not merely better core utilisation. Parallel execution has a memory cost, since each worker holds its own augmented matrix: the sampled peak memory increase reached 16 GB on Dexter, against 2 GB for Vanilla and serial Greedy Boruta.

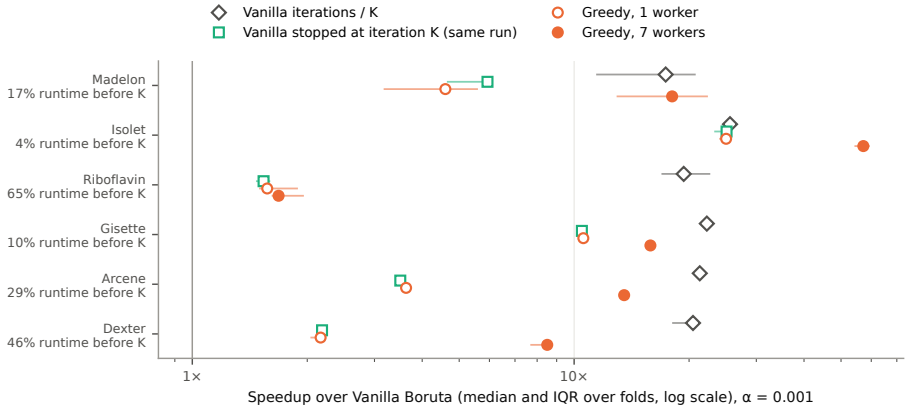


Fig. 2: Speedup over Vanilla Boruta (median and IQR over 15 folds; $\alpha = 10^{-3}$). Diamonds: Vanilla iterations/K; squares: Vanilla's own run stopped at iteration K (three refits); circles: observed serial (open) and parallel (filled, 7 workers) Greedy Boruta. Row labels give the share of Vanilla's runtime spent before iteration K.

4.3 Selection Quality Against Ground Truth

Recall. Greedy and Vanilla Boruta recover all 25 relevant features in every fold of the reference and $p = 2{,}000$ scenarios. With weak signal, both find about a third of them (Greedy: 0.34, Vanilla rough fix: 0.34, Vanilla strict: 0.32 at $\alpha = 10^{-3}$). In the coupled runs, Vanilla's strict and rough-fix selections are subsets of Greedy Boruta's in all 495 fold–α combinations, as Theorem **??** requires.

False positives. Fig. **??** compares the mean number of false positives per fold with the bound $m_0 K/(p+1)$ of Theorem **??**. Greedy Boruta stays at 22–50% of the bound in every scenario and at every α, and the false positives grow much more slowly than K. The per-feature false positive rate falls from about 1.1% at $p = 500$ to 0.45% at $p = 2{,}000$, as the bound predicts. The gap to Vanilla Boruta is far smaller than the idealised comparison in Appendix **??** suggests. In the null scenario, where every selection is false, Vanilla confirms 4.7 (strict) and 5.0 (rough fix) noise features per fold against Greedy Boruta's 5.5; only with strong signal (reference) does Vanilla select noticeably fewer (1.7 strict, 2.9 rough fix, against 5.0). The reason is that the data are fixed across iterations: a noise feature with a chance association to the target in the training sample beats the shadows repeatedly, violating the independence across iterations behind Vanilla's near-zero theoretical false positive rate. Such features are confirmed by both methods; Vanilla merely needs more iterations to do so.

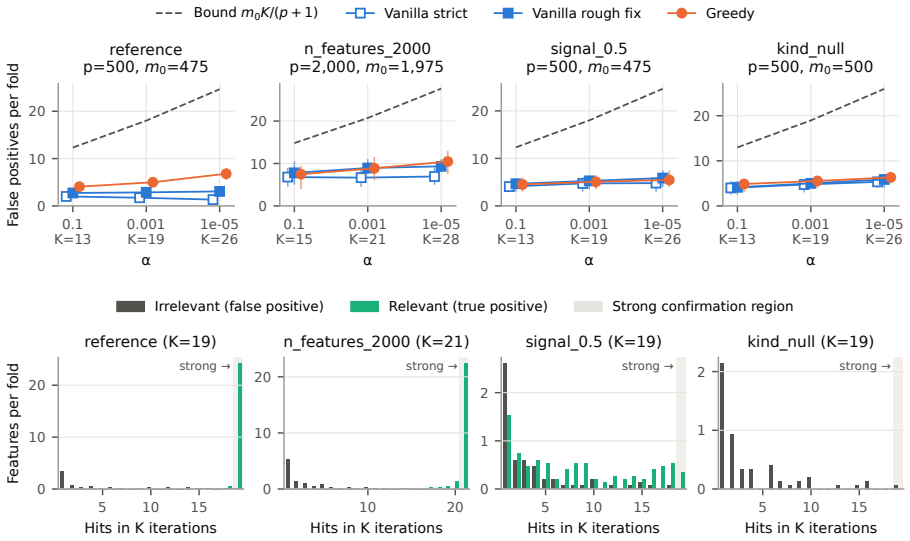


Fig. 3: Top: false positives per fold against α (mean and IQR) with the bound $m_0K/(p+1)$. Bottom: hits within K iterations of the features selected by coupled Greedy Boruta ($\alpha = 10^{-3}$), per fold; the shaded region is the strong-confirmation range.

Weak and strong confirmations. The hit counts in Fig. **??** (bottom) separate the two kinds of selection. With strong signal, relevant features obtain a hit in essentially every iteration and are strongly confirmed, whereas false positives typically obtain one or two hits (63% have at most two). Across scenarios and α, 98–100% of Greedy Boruta's false positives are weak confirmations. With weak signal, relevant features also spread over low hit counts, and the weak set contains true and false positives in comparable numbers, which is exactly the population that the weak label flags for further scrutiny.

4.4 Real-World Benchmarks

What Greedy Boruta adds. Fig. **??** decomposes each coupled Greedy selection by Vanilla Boruta's outcome. Vanilla's final selection makes up 84–97% of Greedy Boruta's on four datasets, 76% on Gisette and 45% on Arcene. Strong confirmations, which Vanilla makes at iteration K, range from 87% of the selection (Isolet) to almost none (Arcene, with 160 training samples). The remaining weak confirmations are mostly features that Vanilla Boruta also confirms, only later. In the refitted runs, Vanilla confirmed 66–71% of Greedy's weak features on Madelon, Isolet, Riboflavin and Dexter, and decided them at a median iteration of 39–74, two to three times K. On Gisette and Arcene, about half of the weak features are eventually rejected. These are the features that make Greedy Boruta's selection larger (Gisette: 2,035 against 1,547; Arcene: 420 against 187).

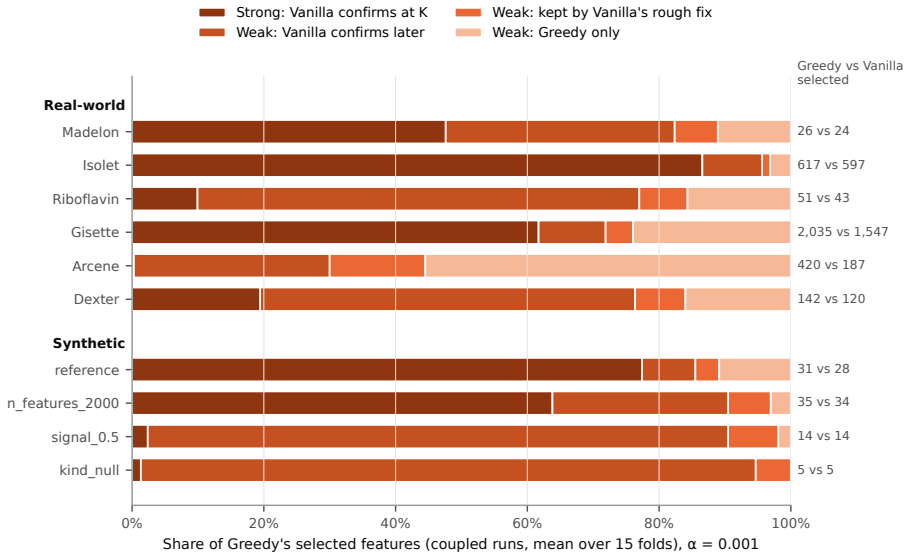


Fig. 4: Composition of coupled Greedy Boruta selections by Vanilla Boruta's outcome ($\alpha = 10^{-3}$, mean over 15 folds). Strong confirmations are exactly Vanilla's confirmations at iteration K; the other layers are weak confirmations that Vanilla confirms later, keeps through the rough fix, or does not select.

Downstream performance. Table **??** and Fig. **??** report the paired comparison. The mean score difference lies within the 0.01 margin, or favours Greedy Boruta, in 17 of the 18 dataset–α combinations; the exception is Riboflavin at $\alpha = 10^{-5}$ (-0.012 in R^2, with a fold standard deviation of about 0.2). Non-inferiority is established on Gisette and Isolet at every α and is inconclusive on the other four datasets, also at the sensitivity level 0.025. This reflects the resolution of the test rather than a difference between the methods: with 15 overlapping folds the corrected standard error is three to five times the margin on the small datasets (Arcene, Riboflavin), and the all-feature model, which performs no selection at all, passes or fails in exactly the same places. Coupling the random streams reduces the standard deviation of the paired differences by 10–50%. Finally, α has little influence beyond the iteration budget (Fig. **??**): smaller α lowers the speedup moderately through a larger K, while selection sizes and score differences barely change.

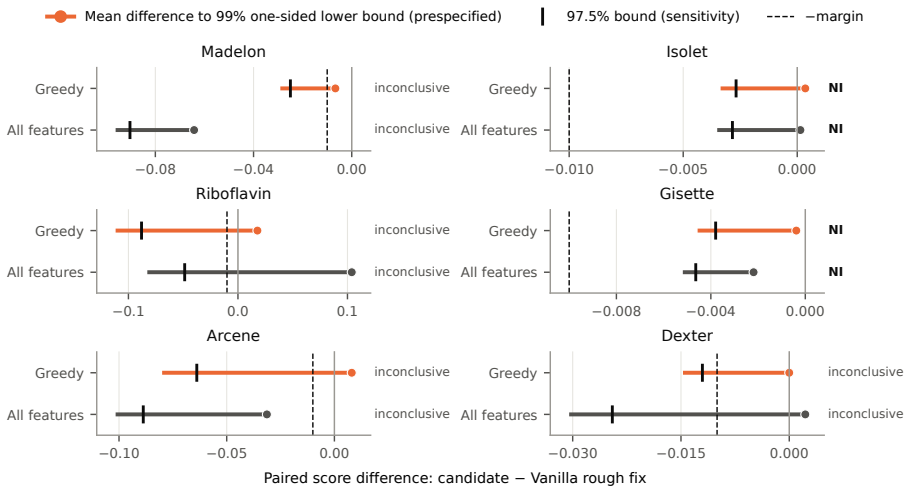


Fig. 5: Paired score difference with Vanilla Boruta ($\alpha = 10^{-3}$): mean (dot), prespecified 99% one-sided lower bound (end of bar), 97.5% sensitivity bound (tick) and non-inferiority margin (dashed). The all-feature model calibrates the test.

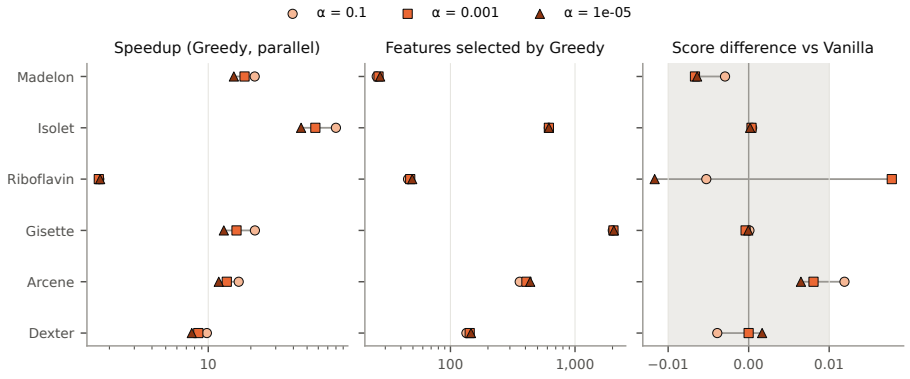


Fig. 6: Sensitivity to α: parallel speedup, number of selected features and score difference with Vanilla Boruta (shaded: non-inferiority margin), one marker per α.

5 Discussion

5.1 The Price of the Single-Hit Rule in Practice

Under idealised assumptions, Greedy Boruta exchanges a vanishing false positive rate for one bounded by $K/(p+1)$ (Theorem **??**, Appendix **??**). The experiments show that the conservatism Vanilla Boruta buys is much smaller in practice. Its near-zero rate assumes that a noise feature's hits are independent across iterations, but all iterations share the same training data, so a feature with a chance association in the sample keeps beating the shadows. Vanilla then confirms it too, only later: in the null scenario it confirms 95% of the noise features that Greedy Boruta selects. Greedy Boruta's additional false positives are few, remain well below the bound, decrease with p, and are almost all weak confirmations with one or two hits.

What Vanilla Boruta's extra iterations do buy is a decision on the small set of borderline features left at iteration K. Most of these are eventually confirmed; on Gisette and Arcene, about half are rejected, and there Greedy Boruta's selection is markedly larger. Combined with the weak/strong split, this suggests a practical workflow: use Greedy Boruta to resolve all features quickly, and direct further scrutiny, such as stability analysis, domain review or a longer Vanilla run restricted to these features, at the weak confirmations only.

5.2 Limitations

1. The FPR bound relies on shadow exchangeability (Assumption **??**). Importance biases (e.g. towards high-cardinality features) affect both methods, but features with occasional, non-significant hits are confirmed weakly by Greedy Boruta and may remain tentative or be rejected by Vanilla Boruta.
2. Speedups depend on how Vanilla Boruta distributes its work. When most features are rejected at iteration K and the remaining iterations are cheap (Riboflavin, Dexter), the gain of serial Greedy Boruta is modest; parallel execution recovers much of it at the cost of memory proportional to the number of workers.
3. The predictive comparison has limited resolution on small datasets: with 15 overlapping folds, non-inferiority at a 0.01 margin could not be established even for the all-feature model. The synthetic results come from a single realisation of each scenario.
4. Both methods inherit the limitations of the base importance measure. With a pure XOR target, whose two relevant features are marginally uninformative, neither method nor the downstream model on all features detects the signal (Appendix **??**).

6 Conclusion

We have presented Greedy Boruta, a modification to the Boruta algorithm that replaces the binomial-test confirmation criterion with a single-hit rule: a

feature is confirmed if it exceeds the maximum shadow importance in at least one iteration. This modification yields four formal guarantees: a conditional feature recall guarantee relative to Vanilla Boruta (Theorem **??**), convergence in exactly $K = \lceil \log_2(p/\alpha) \rceil$ iterations under Bonferroni correction (Theorem **??**), a per-feature false positive rate bounded above by $K/(p+1)$ (Theorem **??**) and the simplification of the final BH rejection calculation (Corollary **??**). As a consequence, Greedy Boruta eliminates all termination-related hyperparameters and substantially simplifies the multiple-testing correction framework used by Boruta.

Beyond the core algorithm, we introduced two practical extensions. First, we proposed a weak/strong confirmation framework that distinguishes features satisfying the original Boruta confirmation criterion from those selected exclusively by the greedy criterion, providing users with an interpretable measure of confidence among the confirmed features. Second, in Appendix **??**, we showed that the number of shadow features can be reduced from p to $r \leq p$, yielding an explicit trade-off between computational efficiency and false positive control. This extension is particularly attractive in high-dimensional and distributed computing environments, where the creation and storage of shadow features can become a substantial computational bottleneck.

Experiments on a synthetic benchmark with known ground truth and six real-world high-dimensional datasets support the theoretical analysis and demonstrate substantial computational gains, with speedups ranging from 1.8 to 66.2× while maintaining downstream predictive performance that is not detectably different from Vanilla Boruta in most configurations under the reported tests; this does not establish predictive equivalence.

We recommend Greedy Boruta as a practical default for all-relevant feature selection when downstream filtering, regularization or model validation procedures are available. The method is particularly appealing in high-dimensional settings, where its false positive rate remains small while its computational advantages become most pronounced. Future work includes extending the greedy criterion to alternative feature-importance measures, such as SHAP and permutation importance, and investigating the statistical properties of weakly confirmed features in large-scale applications. An open-source implementation is publicly available through the `greedyboruta` package on PyPI.

A Proofs

For each feature j, define the total number of hits across K iterations as

$$H_j = \sum_{i=1}^{K} \mathbf{1}\left\{ I_j^{(i)} > M^{(i)} \right\},$$

where $I_j^{(i)}$ denotes the importance of feature j at iteration i and $M^{(i)} = \max_j S_j^{(i)}$ denotes the maximum shadow importance at iteration i.

A.1 Proof of Theorem ?? (Recall Guarantee)

Proof. Let $H_j(K)$ count hits through Greedy Boruta's terminal iteration K. If $H_j(K) = 0$, feature j cannot have been confirmed by a binomial test through K, since its confirmation p-value is 1. Since Vanilla Boruta applies the same rejection test at iteration K, its rejection p-value there is $2^{-K} \leq \alpha/p$ under Bonferroni correction (or $2^{-K} \leq \alpha$ without it). Vanilla therefore rejects it by K if it has not already been rejected.

Because rejections are final, feature j cannot be confirmed later, either by a binomial test or by the rough fix, which only considers tentative features. Consequently, every feature confirmed at Vanilla's terminal iteration $T \geq K$ must have $H_j(K) \geq 1$. The shared importance trajectory ensures that Greedy Boruta also observes that hit and selects j at K. Therefore $\mathcal{C}_V(T) \subseteq \mathcal{C}_G(K)$; the terminal iteration counts need not be equal. □

A.2 Proof of Theorem ?? (Convergence Guarantee)

Proof. At iteration K, partition the features into

$$\mathcal{H}^+ = \{j : H_j \geq 1\} \quad \text{and} \quad \mathcal{H}^0 = \{j : H_j = 0\}.$$

By Algorithm **??** (line 14), all features in $\mathcal{H}^+$ are confirmed by Greedy Boruta. In contrast, for any feature $j \in \mathcal{H}^0$, the rejection test evaluates

$$H_0 : p_j \geq 0.5 \quad \text{versus} \quad H_1 : p_j < 0.5,$$

where p_j denotes the probability that feature j exceeds the maximum shadow importance in a given iteration. Under the null hypothesis, the hit count follows

$$H_j \sim \mathsf{Binomial}(K, \theta = 0.5).$$

Since $H_j = 0$, the rejection p-value is

$$\mathbb{P}(H_j = 0 \mid K, \theta = 0.5) = (0.5)^K = 2^{-K}.$$

Notice that all features in $\mathcal{H}^0$ have this identical p-value. Under Bonferroni correction across $|\mathcal{H}^0| \leq p$ features, rejection requires $2^{-K} \leq \alpha/p$. The minimum integer satisfying this is $K = \lceil \log_2(p/\alpha) \rceil$. In practice, at iteration K all features in $\mathcal{H}^0$ are rejected and all in $\mathcal{H}^+$ are confirmed, so the algorithm terminates with no tentative features. □

Note: Without Bonferroni correction, rejection requires $2^{-K} \leq \alpha$, giving $K = \lceil \log_2(1/\alpha) \rceil$.

A.3 Proof of Corollary ?? (BH Rejection Simplification)

Proof. From the proof of Theorem **??**, any tentative feature j at iteration K has $H_j = 0$ and p-value $p^* = 2^{-K}$.

The Benjamini-Hochberg procedure orders p-values $p_{(1)} \leq \cdots \leq p_{(m)}$ and finds the largest i such that $p_{(i)} \leq (i/m) \cdot \alpha$. Since all p-values are identical ($p_{(i)} = p^*$ for all i), this reduces to checking whether $p^* \leq \alpha$. Under either correction setting, $K \geq \lceil \log_2(1/\alpha) \rceil$, so $p^* = 2^{-K} \leq \alpha$ and the condition holds at $i = m$ and all tentative features are rejected. Hence, the FDR-corrected and uncorrected procedures coincide. □

A.4 Proof of Theorem ?? (FPR of Greedy Boruta)

Proof. For an irrelevant feature j, let $A_i = \{I_j^{(i)} > M^{(i)}\}$ denote a hit in iteration i. Under Assumption **??**, each of the $p+1$ importances has the same probability of being the unique maximum. These events are disjoint, so $\mathbb{P}(A_i) \leq 1/(p+1)$, with equality if the maximum is unique almost surely. A false confirmation occurs if at least one A_i occurs. The union bound gives

$$\mathbb{P}\left(\bigcup_{i=1}^{K} A_i\right) \leq \sum_{i=1}^{K} \mathbb{P}(A_i) \leq \frac{K}{p+1}.$$

This bound allows ties and does not require independence across iterations. Under the additional assumptions of independent hit events across iterations and an almost surely unique maximum in every iteration, the exact probability is $1 - (p/(p+1))^K$, which the bound above dominates.

For the expected number of false positives, let $\mathcal{N}$ be the set of m_0 irrelevant features. By linearity of expectation, which holds for dependent random variables,

$$\mathbb{E}[\text{FP}] = \sum_{j \in \mathcal{N}} \mathbb{P}(j \text{ is confirmed}) \leq \frac{m_0 K}{p+1}.$$

□

Comparison with Vanilla Boruta. Under Assumption **??**, the FPR of Vanilla Boruta is effectively negligible. In Vanilla Boruta, without Bonferroni correction, a feature is confirmed only if its hit count exceeds a threshold h^* satisfying

$$\mathbb{P}(\mathsf{Binomial}(K, \theta = 0.5) \geq h^*) \leq \alpha.$$

For the following idealized fixed-horizon comparison, assume independent hit events across iterations and an almost surely unique maximum in addition to exchangeability. Then the per-iteration hit probability is $\theta_0 = 1/(p+1)$, and its hit count follows

$$H_j \sim \mathsf{Binomial}(K, \theta_0).$$

Consequently, the probability that an irrelevant feature is falsely confirmed by Vanilla Boruta is $\mathbb{P}(H_j \geq h^*)$, which decays exponentially in K and is therefore

extremely small in high-dimensional settings. For example, with $\alpha = 0.001$ and $p = 500$, Greedy Boruta requires $K = 19$ iterations, while Vanilla Boruta requires at least $h^* = 19$ hits for confirmation. Under the null hit probability $\theta_0 = 1/501$, the resulting false positive probability for Vanilla Boruta is approximately 10^{-52}, compared to the false positive probability under Greedy Boruta

$$\mathbb{P}(H_j \geq 1) = 1 - \left(\frac{500}{501}\right)^{19} \approx 3.73\%.$$

This contrast reflects the central trade-off between the two procedures: Greedy Boruta sacrifices the near-zero FPR of Vanilla Boruta in exchange for deterministic convergence, elimination of tentative features, and substantially improved computational efficiency.

B Reducing the Number of Shadow Features

By relaxing the confirmation criterion for relevant features while preserving the statistical rigour of the rejection criterion, Greedy Boruta achieves substantial reductions in computational time across the datasets considered in Section **??**. A key advantage of Greedy Boruta is that the total number of iterations can be determined analytically from the significance level α and the number of features p, unlike Vanilla Boruta, whose convergence behavior depends on sequential hypothesis testing and heuristic stopping rules. In this section, we further investigate how the computational cost of Greedy Boruta can be reduced by decreasing the number of shadow features used to define the random-noise threshold at each iteration. We refer to this variant as *reduced-shadow* Greedy Boruta.

Recall that both Vanilla Boruta and Greedy Boruta augment the original dataset containing p features with p shadow features obtained through random permutations of the original covariates. Consequently, as shown in Algorithm **??**, the base estimator is trained on a design matrix of dimension $n \times 2p$. However, when p is large, constructing all p shadow features may become computationally expensive. To address this, we consider using only $r \leq p$ shadow features instead of the full set of p shadows. In practice, at each iteration, we generate only r shadow features obtained by randomly permuting r selected covariates. The random-noise threshold is then defined as the maximum importance among the generated shadow features.

This modification is particularly attractive in high-dimensional settings, where the cost of fitting the base estimator dominates the overall runtime. Moreover, reducing the number of shadow features can substantially decrease memory and storage requirements. This is especially relevant in distributed computing environments for big data, such as Apache Spark [**?**], where the augmented dataset may not fit entirely in memory. In such settings, shadow features either need to be represented explicitly in the execution plan or materialized as intermediate data on disk, both of which introduce additional computational and storage overhead. Using fewer shadow features alleviates these costs while preserving the statistical framework of the method.

Let $\theta_0^\star$ denote the per-iteration hit probability of an irrelevant feature when only r shadow features are used. Under Assumption **??** for these $r+1$ importances,

$$\theta_0^\star \leq \frac{1}{r+1},$$

with equality when the maximum is unique almost surely. Applying the union-bound argument from Theorem **??** gives

$$\begin{aligned}\mathbb{P}(H_j \geq 1) &\leq \sum_{i=1}^{K} \mathbb{P}\Big(I_j^{(i)} > M_r^{(i)}\Big) \\ &\leq \frac{K}{r+1}, \end{aligned} \tag{2}$$

where $M_r^{(i)}$ is the maximum importance among the r shadows in iteration i. With the additional assumptions of independent hit events across iterations and an almost surely unique maximum, the exact probability is $1-(1-1/(r+1))^K$. This result highlights a direct trade-off between computational cost and false positive control. Reducing the number of shadow features decreases the dimensionality of the augmented design matrix from $2p$ to $p+r$, potentially yielding substantial computational savings. However, the upper bound on the null hit probability increases from $1/(p+1)$ to $1/(r+1)$, weakening the FPR guarantee.

Under these additional assumptions, Figure **??** illustrates this trade-off for different combinations of number of features p, significance level α and fraction of shadow features r/p. The impact of reducing the number of shadow features depends strongly on the dimensionality of the problem. For small values of p, the FPR is already relatively large when all shadow features are used, so further reductions in r may lead to an undesirable increase in spurious confirmations. For example, when $p = 500$ and $\alpha = 0.05$, the per-feature FPR is approximately 2.8% when $r = p$ and increases 4.6 times to 13% when only 20% of the shadow features are used. In contrast, for moderate and large values of p, the FPR remains small even when only a fraction of the shadow features is used. For instance, when $p = 10{,}000$ and $\alpha = 0.05$, the FPR is approximately 0.18% when $r = p$ and remains below 1% when only 20% of the shadow features are used. In practice, an FPR of γ indicates that a truly irrelevant feature is incorrectly confirmed with probability γ. Therefore, if there are m_0 irrelevant features, the expected number of false positives is $m_0\gamma$.

A practical question is how to choose the number of shadow features r. Given the upper bound on the FPR in Equation (**??**), one could, in principle, select r so that the upper bound does not exceed a significance level α. Solving

$$\frac{K}{r+1} \leq \alpha$$

yields the condition $r \geq K/\alpha - 1$. Therefore, define the minimum required shadow count as

$$r_{\min} = \left\lceil \frac{K}{\alpha} - 1 \right\rceil.$$

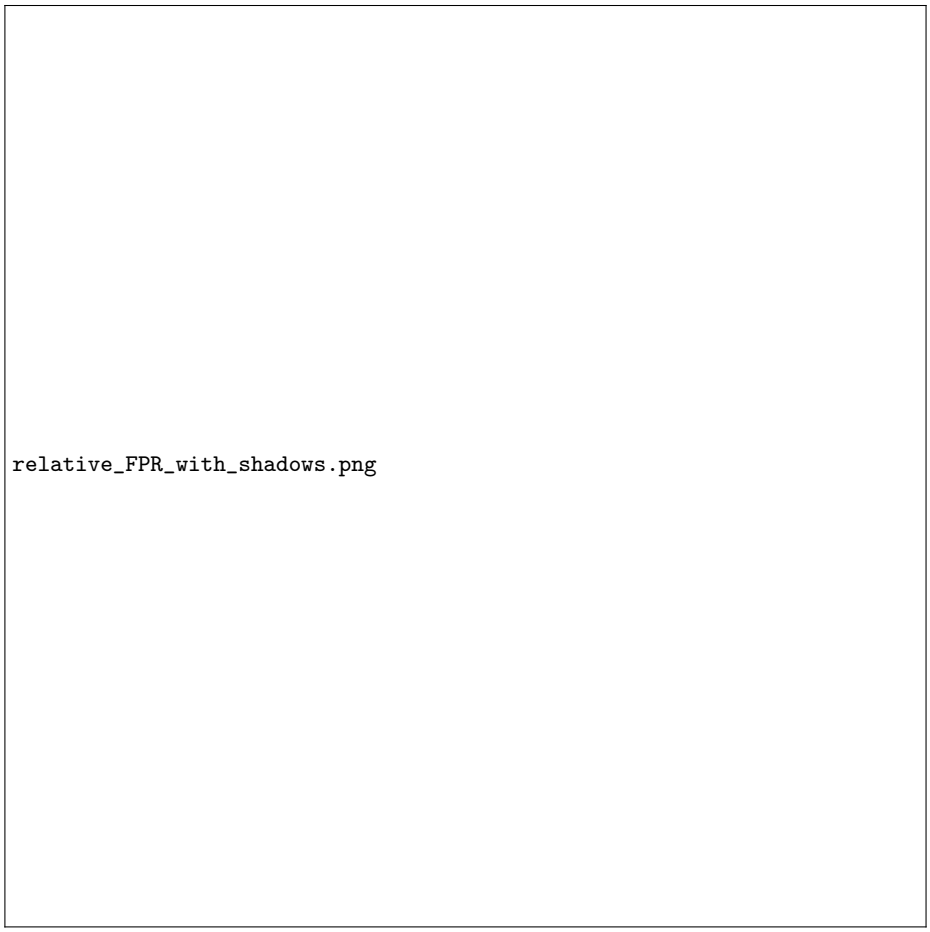


Fig. 7: Per-feature false positive rate (FPR) relative to the minimum FPR across all α at 100% of Greedy Boruta under Bonferroni correction using different number of features, fraction of shadow features and significance level.

Choose $r = r_{\min}$ only when $1 \leq r_{\min} \leq p$ (or $r = 1$ if $r_{\min} < 1$). If $r_{\min} > p$, no $r \leq p$ can certify the target using this bound. Clipping $r_{\min}$ to p does not preserve the guarantee. This is a feasibility limitation of the bound, not a proof that the actual FPR exceeds the target.

B.1 Reduced-Shadow Greedy Boruta on Real-World Benchmarks